EXPERIMENT-23:

AIM:

To write a C program to encrypt and decrypt plaintext using Simplified DES (S-DES) in Counter (CTR) mode with an initial counter value of `00000000`.

CODE:

```
#include <stdio.h>


int IP[]={2,6,3,1,4,8,5,7}, IP1[]={4,1,3,5,7,2,8,6};
int EP[]={4,1,2,3,2,3,4,1}, P4[]={2,4,3,1};
int P10[]={3,5,2,7,4,10,1,9,8,6}, P8[]={6,3,7,4,8,5,10,9};
int S0[4][4]={{1,0,3,2},{3,2,1,0},{0,2,1,3},{3,1,3,2}};
int S1[4][4]={{0,1,2,3},{2,0,1,3},{3,0,1,0},{2,1,0,3}};
int k1,k2;

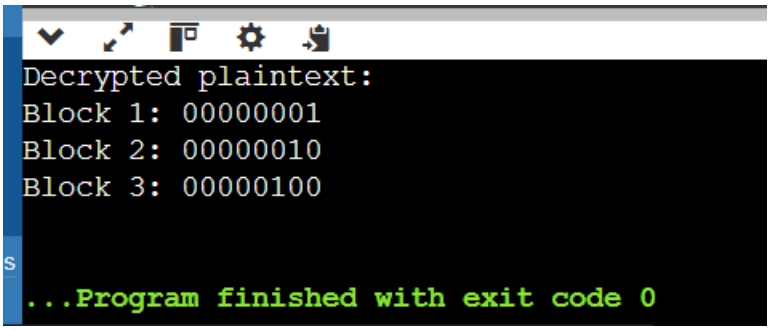

int perm(int x,int *p,int n,int bits){
   int y=0,i;
   for(i=0;i<n;i++) y=(y<<1)|((x>>(bits-p[i]))&1);
   return y;
}
int rot(int x,int n){return ((x<<n)|(x>>(5-n)))&31;}


void keys(int k){
   int t=perm(k,P10,10,10),l=t>>5,r=t&31;
   l=rot(l,1);r=rot(r,1);k1=perm((l<<5)|r,P8,8,10);
   l=rot(l,2);r=rot(r,2);k2=perm((l<<5)|r,P8,8,10);
}
int fk(int x,int k){
```

```
    int l=x>>4,r=x&15,e=perm(r,EP,8,4)^k;
    int a=e>>4,b=e&15;
    int s0=S0[((a&8)>>2)|(a&1)][(a>>1)&3];
    int s1=S1[((b&8)>>2)|(b&1)][(b>>1)&3];
    return (((l^perm((s0<<2)|s1,P4,4,4))<<4)|r);
}
int enc(int x){
    x=perm(x,IP,8,8);x=fk(x,k1);
    x=((x&15)<<4)|(x>>4);x=fk(x,k2);
    return perm(x,IP1,8,8);
}
void bin(int x){int i;for(i=7;i>=0;i--)printf("%d",(x>>i)&1);}

int main(){
    int p[]={1,2,4},c[3],i;
    keys(0xFD);
    printf("CTR Mode using S-DES\nCounter starts at 00000000\n");
    for(i=0;i<3;i++){
        c[i]=p[i]^enc(i);
        printf("Cipher block %d: ",i+1);bin(c[i]);printf("\n");
    }
    printf("Decrypted plaintext:\n");
    for(i=0;i<3;i++){
        printf("Block %d: ",i+1);bin(c[i]^enc(i));printf("\n");
    }
    return 0;
}
```

OUTPUT:


Decrypted plaintext:
Block 1: 00000001
Block 2: 00000010
Block 3: 00000100
...Program finished with exit code 0